Mark schemes

Q1.

$$\frac{(2x+3)(x-1)}{(x+1)(x-1)}$$

numerator / *denominator* } *not necessarily in fraction*

B1
B1

$$= \frac{2x+3}{x+1}$$

CAO in this form. Not $\frac{2x+3}{x+1}\frac{\cancel{x-1}}{\cancel{x-1}}$

B1

3

Alternative

$$\frac{2x^2 - 2 + x - 1}{x^2 - 1}$$

$$= 2 + \frac{x-1}{x^2-1}$$

(M1)

$$= 2\ \frac{x-1}{(x-1)(x+1)}$$

(B1)

$$= 2 + \frac{1}{x+1}$$

(A1)

[3]

Q2.

(a) (i) p(2) = 8 + 4 − 20 + 8

Finding p(2) *M0 long division*

M1

$= 0$, $\Rightarrow$ $x - 2$ is a factor

Shown = 0 ***AND*** *conclusion/ statement about $x - 2$ being a factor*

A1 **2**

(ii) Attempt at quadratic factor

or factor theorem again for 2nd factor

M1

$x^2 + 3x - 4$

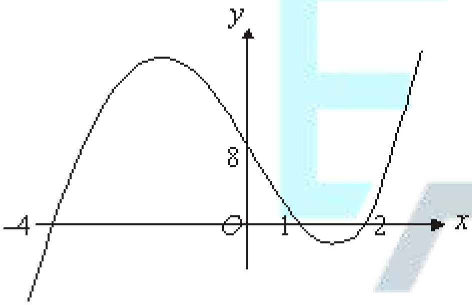
or $(x+4)$ or $(x-1)$ proved to be a factor

A1

$p(x) = (x - 2)(x + 4)(x - 1)$

A1 **3**

(b)



Graph through (0,8) 8 marked

B1

Ft "their factors" 3 roots marked on x-axis

B1ft

Cubic curve through their 3 points

M1

Correct including x-intercepts correct

Condone max on y-axis etc or slightly wrong concavity at ends of graph

A1 **4**

[9]

Q3.

(a) $f(1) = 0$

B1

1

(b) f(−2) = − 24 + 8 + 14 + 2 = 0

B1

1

(c) $\dfrac{(x-1)(x+2)}{3x^3+2x^2-7x+2} = \dfrac{(x-1)(x+2)}{(x-1)(x+2)(ax+b)}$

Recognising (x − 1), (x + 2) as factors PI

B1

$ax^3 = 3x^3$ $-2b = 2$

a

B1

$a = 3$ $b = -1$

b

B1

Or *By division M1 attempt started*
M1 complete division
A1 Correct answers

3

[5]

Q4.

(a) p(3) = 27 − 36 + 9

*Finding p(3) and **not** long division*

M1

p(3) = 0 $\Rightarrow$ $x - 3$ is a factor

Shown = 0 ***plus a statement***

A1

2

(b) $x(x^2 - 4x + 3)$ or $(x - 3)(x^2 - x)$ attempt

Or p(1) = 0 $\Rightarrow$ *x − 1 is a factor attempt*

M1

p(x) = $x(x - 1)(x - 3)$

Condone x + 0 or x − 0 as factor

A1
2
[4]

Q5.

(a) (i) p(2) = 0

B1
1

(ii) See $-\frac{1}{2}$

B1

$$p\left(-\frac{1}{2}\right)=6\times\left(-\frac{1}{8}\right)-19\times\frac{1}{4}+9\left(-\frac{1}{2}\right)+10$$

Use $\pm\frac{1}{2}$

M1

= 0

Arithmetic to show = 0 and conclusion.
Long division : 0/3

A1
3

(iii) $p(x)=(2x+1)(x-2)(3x-5)$

x − 2

B1

Complete expression

B1
2

(b) $\dfrac{3x(x-2)}{(2x+1)(x-2)(3x-5)}$

For $\dfrac{3x(x-2)}{their(a)(iii)}$

M1

$= \dfrac{3x}{(2x+1)(3x-5)}$

Or $\frac{3x}{6x^2 - 7x - 5}$ *No ISW on A1*

A1

[8]

Q6.

(a) (i) $x - 1$ is a factor

May be earned retrospectively

B1

$x + 2$ is a factor

From part (iii)

B1

2

(ii) Attempt at third factor

Multiplying/ dividing/factor theorem

M1

f(x) = (x – 1)(x + 2)(x – 4)

(x+4) $\Rightarrow$ *M1, A0*

A1

2

(b) (i) At A , y = 8

Or (0,8)

B1

1

(ii) At B , x = 4

Or (4,0) NO ft of wrong factor

B1

1

(c) (i) $\frac{x^4}{4} - x^3 - 3x^2 + 8x \quad (+c)$

Increase one power by 1

M1

One term correct (unsimplified)

A1

Two other terms correct (unsimplified)

A1

All correct (unsimplified)
(condone missing + c)

A1

4

(ii) Realisation that limits are –2 and 1

Condone wrong way round

B1

Area = $\left[\frac{1}{4} - 1 - 3 + 8\right] - [4 + 8 - 12 - 6]$

Attempt to sub their limits into their (c)(i)

M1

= $20\frac{1}{4}$

CSO. Must use F(1) – F(–2) correctly

A1

3

[13]

Q7.

(a) $k^2 + 10k + 25 - 12k^2 - 24k$

Condone one slip

M1

$= -11k^2 - 14k + 25$

No ISW here

A1

2

(b) (i) Real roots when "$b^2 - 4ac$" ≥ 0

Non-negative discriminant (stated / used)

B1

$(k + 5)^2 - 12k(k + 2)$

Finding $b^2 - 4ac$ in terms of k
Or factorisation attempt.

M1

$(k - 1)(11k + 25)$ attempted to be shown m1

equal to $11k^2 + 14k - 25$

$-11k^2 - 14k + 25 \geq 0$

$\Rightarrow (k - 1)(11k + 25) \leq 0$

Real roots condition correct and ...
AG (be convinced about inequality)

A1

5

(ii) (Critical values) 1 and $-\dfrac{25}{11}$ seen

B1

Sketch or sign diagram

M1

$\Rightarrow -\dfrac{25}{11} \leq k \leq 1$

A1

3

[10]

Q8.

(a) (i) $(2 + x)^3 =$

M1

$(2^3) + 3(2^2)(x) + 3(2)(x^2) + (x^3)$

A1

... = 8 + 12x + 6x2 + x3 (*)

Any valid method; must contain all components

Accept a = 12

Accept b = 6

A1

3

(ii) $(2 - x)^3 = 8 - 12x + 6x^2 - x^3$ (**)

Clear $x \rightarrow -x$ in (i) OE

M1

ft numerical a and b

A1ft

2

(b) $(2+x)^3 - (2-x)^3$ = (*) – (**)

Subtracts the 2 expressions in (a)

M1

…… = $24x + 2x^3$

CSO AG (be convinced)

A1

2

(c) $\frac{dy}{dx} = 24 + 6x^2$

A power of x decreased by 1

M1

For st. pt. $24 + 6x^2 = 0$

A1

Not possible since $24 + 6x^2 > 0$

Any valid explanation

E1

3

[10]

Q9.

(a) (i) p(3) = 27 – 45 + 21 – 3

Finding p(3)

M1

p(3) = 0 $\Rightarrow$ $x - 3$ is a factor

Shown = 0 plus a statement

Or $(x - 3)(x^2 - 2x + 1)$

A1

2

(ii) B is point (3,0)

Must have coordinates

B1

1

(b) (i) $\frac{dy}{dx} = 3x^2 - 10x + 7$

One term correct

M1

All correct with NO + c etc

A1

2

(ii) $3x^2 - 10x + 7 = 0$

Putting their $\frac{dy}{dx} = 0$

M1

$\Rightarrow (x - 1)(3x - 7) = 0$

Attempt to use quad formula or factorise

m1

$\Rightarrow$ at M, $x = \frac{7}{3}$

***CSO** factors correct etc*

A1

3

(c) $\frac{d^2y}{dx^2} = 6x - 10$

ft their $\frac{dy}{dx}$

B1ft

sub $x = 1$, $\Rightarrow \frac{d^2y}{dx^2} = -4$

ft their $\frac{d^2y}{dx^2}$

B1ft

2

(d) (i) $\frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$ $(+c)$

Increase one power by 1

M1

One term correct

A1

Two other terms correct

A1

All correct (condone missing + c)

A1

4

(ii) Realisation that limits are 0 and 1

Condone wrong way round

B1

$\left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3\right] - 0$

Attempt to sub their limits into their (d)(i)

M1

$= -\frac{11}{12}$

CSO. *Must use F(1) – F(0) correctly*

A1

Area = $\frac{11}{12}$

CSO. *Convincing argument*

E1

4

[18]

Q10.

(a) $p(x) = x^3 + x^2 + 3x - 2x^2 - 2x - 6$

Condone one slip

M1

$= x^3 - x^2 + x - 6$

$a = -1$, $b = 1$

A1

2

(b) Considering $x^2 + x + 3 = 0$ and attempting to solve or use discriminant

$b^2 - 4ac = 1 - 12 = -11$

M1

$b^2 - 4ac < 0 \Rightarrow$ no real roots

CSO

A1

Only real root is 2

$x = 2$

B1

3

[5]